

Software Requirements Specification

Open Clinical Record

Open Clinical Record Internship Project

MVP Scope Revision — September 2026

Document Status: Revised for Mentor Review

Document Item	Value
Product	Open Clinical Record
Document	Software Requirements Specification (SRS)
Version	1.4 — internship MVP scope revision
Status	Revised for mentor review and implementation baseline
MVP scope	Patient Management, Patient Chart, and Appointment Management
Primary users	Clinician/Doctor; Nurse/Clinical Staff; Receptionist/Front Desk
Implementation period	Remaining one-month internship period
Standards	International interoperability standards are deferred from the MVP
Future functionality	Additional EMR capabilities may be considered after the core MVP is complete

This document intentionally limits the internship MVP to functionality that can realistically be implemented, tested, and demonstrated within the remaining project period.

Contents

1	Introduction	4
1.1	Purpose of the Requirements Document	4
1.2	Scope of Product	4
1.3	Definitions and Abbreviations	4
1.4	References	5
1.5	Overview of the Remainder of the Document	6
2	General Description	6
2.1	Product Perspective	6
2.2	Product Functions	6
2.3	User Characteristics	7
2.4	General Constraints	7
2.5	Assumptions and Dependencies	8
3	Specific Requirements	8
3.1	External Interface Requirements	8
3.1.1	User Interface Requirements	8
3.1.2	Software Interfaces	9
3.1.3	Communication Interfaces	9
3.2	Functional Requirements	9
3.2.1	Patient Management Requirements	9
3.2.2	Appointment and Visit Requirements	10
3.2.3	Encounter and Clinical Assessment Requirements	11
3.2.4	Medical Records and Reports Requirements	11
3.2.5	Authentication, Authorization and Audit Requirements	12
3.2.6	Validation and Error Handling Requirements	12
3.3	Business Rules	13
3.4	Non-Functional Requirements	13
3.4.1	Security and Privacy	13
3.4.2	Data Integrity and Clinical Record Safety	14
3.4.3	Performance Requirements	14
3.4.4	Reliability and Failure Handling	15
3.4.5	Usability and Accessibility	15
3.4.6	Maintainability and Extensibility	15
3.4.7	Interoperability and FHIR Readiness	16
3.4.8	Auditability	16

3.4.9	Backup and Recovery	16
3.5	Logical Database Requirements	17
3.6	Design Constraints	18
3.7	System Attributes and Quality Characteristics	18
4	Appendices	19
4.1	Appendix A — Data Structure Specification	19
4.2	Appendix B — Data-Flow and Workflow Models	19
4.3	Appendix C — Detailed Domain/Object Model	20
4.4	Appendix D — Role and Permission Matrix	20
4.5	Appendix E — Requirements Traceability Summary	21
4.6	Appendix F — Clinical Validation Evidence	22
4.7	Appendix G — Open Questions and Items for Mentor Review	22
4.8	Appendix H — Requirements Change Control	23
5	Index	23

1 Introduction

1.1 Purpose

This Software Requirements Specification defines the reduced and prioritized MVP for Open Clinical Record (OCR). The scope has been revised following mentor feedback to fit the remaining one-month internship period.

The MVP is focused on three core business areas: **Patient Management**, **Patient Chart**, and **Appointment Management**. The objective is to deliver these workflows as a coherent, working system before considering additional functionality.

1.2 Scope of Product

Open Clinical Record is a focused outpatient patient-management and appointment system with a basic electronic patient chart. The patient is the central record around which the three MVP areas are connected.

**Patient Management → Patient Chart → Appointment Management →
Check-in/Visit History**

The MVP does not attempt to implement a complete hospital EMR. Full clinical encounter documentation, diagnosis, treatment, prescriptions, advanced medical reports, laboratory, pharmacy, billing, radiology, patient portal, advanced analytics, and standards-based interoperability are deferred.

1.3 Definitions

Term	Definition
EMR	Electronic Medical Record; a digital record containing patient and healthcare-related information.
MVP	Minimum Viable Product; the smallest planned version that satisfies the approved internship scope.
Patient Management	Functions for registering, identifying, searching, viewing, and maintaining basic patient information.
Patient Chart	A longitudinal view of information associated with a patient, including demographics, allergies, medication/history information, alerts where required, and appointment/visit history.
Appointment	A planned date/time for a patient's scheduled visit.
Appointment status	The current state of an appointment, such as scheduled, checked-in, cancelled, completed, or other approved states.
Visit	An occurrence in which a patient attends the clinic; it may originate from an appointment or a basic walk-in.
Walk-in	A patient visit that begins without a previously scheduled appointment.
RBAC	Role-Based Access Control; access based on the authenticated user's assigned application role.

1.4 References

The requirements are informed by the project's existing research and discovery artifacts, including clinical workflow, patient-chart research, appointment-scheduling research, and the mentor-approved scope direction.

- docs/00-project/project-scope.md
- docs/02-discovery/clinical-workflow.md
- docs/01-research/patient-chart-research.md
- docs/01-research/appointment-scheduling-research.md
- docs/03-requirements/non-functional-requirements.md
- docs/04-architecture/architecture-overview.tex

2 General Description

2.1 Product Perspective

The system is designed around a simple clinic workflow. Staff first identify or register the patient, then access the patient chart, manage an appointment, and record attendance/check-in information. The resulting appointment and visit history remains associated with the patient.

The system should be modular enough to support future expansion, but future capabilities must not delay completion of the three approved MVP areas.

2.2 MVP Modules

1. **Patient Management:** patient registration, unique identifier, search, profile, demographic/contact information, and basic status.
2. **Patient Chart:** patient context, allergies, relevant medication/history information, alerts where required, and appointment/visit history.
3. **Appointment Management:** appointment creation, viewing, rescheduling, cancellation, status, check-in, and basic walk-in support.

Authentication, authorization, validation, error handling, and basic audit logging are cross-cutting support concerns. They are not additional business modules.

2.3 User Characteristics

The MVP has exactly three application roles.

Role	Responsibility	MVP capabilities
Receptionist / Front Desk	Patient registration and scheduling	Register/search patients; maintain permitted demographics; create, view, reschedule and cancel appointments; check in patients; support basic walk-ins.
Nurse / Clinical Staff	Clinical support	Search/view patients and charts; view appointments; support check-in/visit workflow; maintain permitted chart information according to the approved permission matrix.
Clinician / Doctor	Clinical oversight and patient care context	Search/view patient charts; review patient history and appointments; perform authorized clinical/chart updates required by the MVP.

No System Administrator or other application role is included in the MVP.

2.4 Constraints

- The MVP must remain small enough to implement and test within the remaining one-month internship period.
- Only three business modules are committed: Patient Management, Patient Chart, and Appointment Management.
- Exactly three application roles are used: Receptionist/Front Desk, Nurse/Clinical Staff, and Clinician/Doctor.
- International healthcare standards and FHIR integration are not required for the internship MVP.
- Additional functionality may be added only after the core MVP is complete and only if time remains.
- Patient information must be protected from unauthorized access.

3 Specific Requirements

3.1 External Interface Requirements

3.1.1 User Interface

The system shall provide clear screens for authentication, patient search, patient registration, patient profile/chart, appointment management, appointment details, check-in, and basic visit history.

The interface shall show the current patient context clearly and shall not present controls for operations the current user is not authorized to perform.

3.1.2 Software Interface

The React frontend shall communicate with the ASP.NET Core backend through the REST API defined by the architecture. The backend shall enforce validation and authorization independently of the frontend.

3.1.3 Communication

Client/server communication shall use secure transport in the deployed application. Communication or save failures shall be reported clearly and shall not falsely indicate successful persistence.

3.2 Functional Requirements

3.2.1 Patient Management Requirements

ID	Requirement	Detailed specification
FR-PM-001	Register patient	The system shall allow an authorized Receptionist/Front Desk user to create a patient record with the required demographic and contact information.
FR-PM-002	Unique patient ID	The system shall assign or record a unique patient identifier for each patient.
FR-PM-003	Search patient	The system shall allow authorized users to search for a patient using supported identifying information.
FR-PM-004	View patient profile	The system shall display the patient's basic demographic, contact, and status information to authorized users.
FR-PM-005	Update patient information	The system shall allow authorized users to update permitted patient information while preserving the integrity of the patient record.
FR-PM-006	Prevent duplicate patients	The system shall validate identifying information and provide appropriate feedback when a likely duplicate patient is detected.
FR-PM-007	Patient status	The system shall maintain a basic patient status and apply approved restrictions to actions affected by that status.

3.2.2 Patient Chart Requirements

ID	Requirement	Detailed specification
FR-PC-001	Open patient chart	The system shall allow an authorized user to open a patient's chart from a valid patient record.
FR-PC-002	Chart summary	The chart shall provide a concise view of important patient information available within the MVP.

FR-PC-003	Allergy information	The system shall allow appropriately authorized users to record and view allergy information associated with the correct patient.
FR-PC-004	Medication/history information	The system shall allow appropriately authorized users to record and view relevant medication/history information associated with the patient.
FR-PC-005	Patient alerts	The system shall support recording and displaying important patient alerts where required by the approved workflow.
FR-PC-006	Appointment history	The patient chart shall allow authorized users to view the patient's appointment history.
FR-PC-007	Visit/check-in history	The system shall preserve and display basic visit or check-in history associated with the patient.
FR-PC-008	Correct-patient association	The system shall ensure chart information is associated with the intended patient and shall prevent accidental cross-patient updates.

3.2.3 Appointment Management Requirements

ID	Requirement	Detailed specification
FR-AP-001	Create appointment	The system shall allow an authorized Receptionist/Front Desk user to create an appointment for an existing patient.
FR-AP-002	View appointments	Authorized users shall be able to view appointments relevant to their work.
FR-AP-003	Appointment details	Each appointment shall identify the patient, scheduled date/time, purpose/reason where required, and current status.
FR-AP-004	Reschedule appointment	The system shall allow authorized users to change an eligible appointment's scheduled date/time while retaining sufficient history.
FR-AP-005	Cancel appointment	The system shall allow authorized users to cancel an eligible appointment and retain its historical existence.
FR-AP-006	Appointment status	The system shall maintain a clear appointment status throughout its lifecycle.
FR-AP-007	Check-in	The system shall allow Receptionist/Front Desk or Nurse/Clinical Staff to record patient check-in for an eligible appointment.
FR-AP-008	Basic walk-in	The system shall support a basic walk-in path without requiring a fabricated prior appointment.
FR-AP-009	Visit linkage	Check-in/visit information shall remain linked to the correct patient and, when applicable, the originating appointment.
FR-AP-010	Status-based validation	The system shall prevent appointment actions that violate approved patient or appointment status rules.

3.2.4 Authentication, Authorization, Validation and Audit

ID	Requirement	Detailed specification
FR-SEC-001	Authenticate user	The system shall authenticate users before granting access to protected functions or patient information.
FR-SEC-002	Enforce three roles	The system shall enforce permissions for exactly three MVP roles: Receptionist/Front Desk, Nurse/Clinical Staff, and Clinician/Doctor.
FR-SEC-003	Restrict access	The system shall deny operations for which the authenticated user lacks permission.
FR-SEC-004	Audit important actions	The system shall record important patient and appointment actions with actor and timestamp information where required.
FR-VAL-001	Validate data	The system shall validate required fields, formats, identifiers, relationships, and relevant business rules before saving data.
FR-VAL-002	Clear errors	The system shall provide understandable error feedback without exposing unnecessary protected information.
FR-VAL-003	Safe failure	A failed operation shall not silently create partial or inconsistent patient/appointment records.

3.3 Business Rules

ID	Business rule
BR-001	Each patient has one unique patient identifier within the system.
BR-002	Patient chart information must belong to the correct patient.
BR-003	Cancelled appointments remain in history rather than being silently deleted.
BR-004	Rescheduling shall not erase the fact that an earlier appointment state existed.
BR-005	A walk-in may create a visit without a prior appointment.
BR-006	Appointment operations must respect the current patient and appointment status.
BR-007	Chart access does not automatically grant permission to modify every type of chart information.
BR-008	Only the three approved application roles may be assigned to MVP users.
BR-009	Additional EMR functionality is deferred until the three core modules are complete.

3.4 Non-Functional Requirements

The existing detailed NFR document remains supporting material, but the internship MVP prioritizes the following qualities:

- **Usability:** common patient and appointment tasks should be straightforward for clinic staff.

- **Performance:** ordinary patient search and appointment operations should respond promptly under expected internship-scale usage.
- **Security:** authenticated users must only access permitted patient information and operations.
- **Data integrity:** patient and appointment records must not be silently corrupted or lost.
- **Reliability:** failed operations must produce safe, understandable outcomes.
- **Maintainability:** the implementation should keep the three modules separated enough for future extension.

3.5 Logical Database Requirements

The database shall support the three MVP modules without introducing unnecessary enterprise complexity.

Patient Management: Patient.

Patient Chart: Allergy, Medication History, Patient Alert, and patient-linked appointment/visit history.

Appointment Management: Appointment, Appointment History/Event where required, Visit.

Cross-cutting: User, Audit Event where implemented.

The central relationships are Patient 1- M Appointment, Patient 1- M Visit, and Appointment 0..1-1 Visit where a visit originated from an appointment. A walk-in visit shall be allowed without an appointment reference.

The final ERD should use the simplest normalized model that supports the approved MVP and can be implemented within the remaining project period.

3.6 Deferred Functionality

The following are explicitly outside the committed MVP and shall not be used to expand the implementation schedule:

- Full encounter documentation
- Diagnosis and treatment documentation
- Prescription management
- Advanced medical report/document lifecycle
- Document amendments/versioning
- Laboratory, pharmacy, billing, insurance, and radiology modules
- Patient portal/mobile application
- SMS and external integrations
- FHIR or other international standards implementation
- Advanced analytics and enterprise scheduling
- AI clinical decision support

These may be considered as future work only after the MVP is functional and tested.

3.7 Acceptance Criteria

The MVP is considered ready for demonstration when the following can be completed successfully:

1. A receptionist can register a patient and receive a unique patient ID.
2. An authorized user can search for and open that patient.
3. The patient's chart displays the stored patient information and relevant history.
4. Authorized staff can record and view supported chart information.
5. A receptionist can create an appointment for the patient.
6. Authorized staff can view the appointment and update its status/check in the patient.
7. An appointment can be rescheduled or cancelled without silently deleting its history.
8. A basic walk-in can be recorded without inventing a previous appointment.
9. Role restrictions prevent unauthorized operations.
10. Invalid input and failed operations are handled without creating inconsistent records.

3.8 Traceability Summary

MVP area	Main requirements
Patient Management	FR-PM-001 through FR-PM-007
Patient Chart	FR-PC-001 through FR-PC-008
Appointment Management	FR-AP-001 through FR-AP-010
Cross-cutting controls	FR-SEC-001 through FR-SEC-004; FR-VAL-001 through FR-VAL-003

3.9 Mentor Review Items

The following decisions should be confirmed before implementation is considered frozen:

- Mandatory patient registration fields.
- Exact permissions for each of the three roles.
- Required allergy, medication/history, and alert fields.
- Appointment statuses required for the MVP.
- Whether basic walk-in support is required for the final demonstration or can be deferred.
- Required level of audit logging.

4 Scope Change Control

Any new feature proposed during implementation should first be classified as either MVP or Future. A Future feature shall not displace unfinished core functionality unless the mentor explicitly approves the scope change.